Final Project Guidelines

115-1 Introduction to Machine Learning

1. Project Objectives

The final project asks each team to complete a sound machine learning workflow on a realistic problem: problem formulation, data preparation, model building, evaluation, interpretation, and communication. A correctly implemented baseline with appropriate evaluation and meaningful analysis is sufficient for the standard requirement; advanced exploration is optional.

2. Project Tracks

Level 1 — Instructor-Provided Challenge (Standard Track)

Choose one instructor-provided challenge. Each challenge will specify a problem, suggested dataset, baseline, and evaluation metric. Complete the baseline correctly, use an appropriate train/validation/test procedure, evaluate with suitable metrics, and explain the main results and failure cases. No novel model or SOTA result is required.

Level 2 — Independent Proposal

Propose your own ML problem and dataset. Define a clear application, justify the dataset and baseline, and explain why the selected methods are appropriate. Any suitable model family may be used, including methods beyond the course scope (e.g., LLMs, diffusion models, RL, GNNs, or agents).

Level 3 — Advanced / Bonus Project

Meet the Level 2 requirements and add a substantial extension: independently learn a new ML method, introduce a technically meaningful idea, improve on a baseline/reference method when feasible, perform deeper robustness or failure analysis, and/or build a useful demo. Level 2 or 3 does not automatically guarantee bonus points; bonus depends on quality, difficulty, and execution.

3. Representative Challenge Options

- ML Consultant / AI for Good — Solve a concrete business, organizational, or social problem (e.g., healthcare, finance, e-commerce, air pollution, traffic, fraud, energy, agriculture).
- Fake Image Detector — Real vs. AI-generated images; baseline: CNN/ResNet18; extensions: ViT, generator-specific analysis, failure analysis.
- Adversarial Attack — Study adversarial examples; baseline: FGSM; extensions: PGD, defense methods, attack vs. defense.
- Medical Diagnosis — Predict disease/health risk from tabular clinical data; baseline: Logistic Regression/Random Forest; extensions: XGBoost, SHAP.
- Object Detection / Recognition — Image recognition or detection on a manageable dataset; baseline: standard classifier or YOLOv8n; extensions: fine-tuning, robustness/error analysis.
- Speech Command Recognition — Google Speech Commands or a similar audio dataset; baseline: CNN; extension: Transformer or robustness analysis.
- Recommendation System — MovieLens or similar data; baseline: collaborative filtering; extension: neural CF or cold-start analysis.

Paired challenge variants may include Generator vs. Detector, Attack vs. Defense, Compression vs. Recognition, Caption vs. Verification, Recommendation vs. Cold Start, and Segmentation vs. Noisy Labels.

Suggested Datasets

Heart Disease — Binary Classification | Adult Census Income — Binary Classification
Air Quality — Regression | Bank Marketing — Binary Classification

Cassava Leaf Disease Classification — Multi-class Image Classification | Taiwan PM2.5 / Air Quality
Diabetes Health Indicators Dataset | Default Payments of Credit Card
Accident Severity | Credit Card Fraud Detection*
*For Credit Card Fraud Detection, accuracy alone is not sufficient; include metrics such as Precision, Recall, F1-score, ROC-AUC, or PR-AUC.

4. General Project Requirements

- Problem & Dataset — Clearly state the problem and why it matters. Describe the source, size, input type, target/labels, and learning task. Self-chosen datasets must be suitable and manageable.
- Preprocessing & Representation — Explain relevant cleaning, missing-value handling, normalization, splitting, feature engineering/transformation, or augmentation. Do not treat preprocessing as a black box.
- Model — Implement the required baseline. Additional models are encouraged only when they provide a meaningful comparison. Explain major model choices and key hyperparameters; exhaustive tuning is not required.
- Evaluation — Use metrics appropriate to the task. Classification: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC. Regression: MSE, RMSE, MAE, R^2. For imbalanced tasks (e.g., fraud detection), accuracy alone is not sufficient.
- Results & Discussion — Report quantitative results, explain which method works best and why, discuss important errors/failure cases/limitations, and suggest realistic improvements.

5. Deliverables

- Project Proposal — 1 page: team members, project track/title, problem, dataset, baseline, evaluation metric, and planned extension (if any).
- Written Report — 5–8 page PDF including figures/tables and references as appropriate.
- Final Presentation — 10 minutes + 5 minutes Q&A per team.
- Optional Demo — Live or recorded demo when it helps communicate the contribution.

6. Grading

Problem Definition & Dataset Understanding 10% | Preprocessing & Feature Engineering 20% | Model Design & Baseline Implementation 30% | Evaluation, Results & Discussion 20% | Report & Presentation 20%.
Bonus: up to +10% for meaningful independent exploration, a strong Level 2/3 extension, substantial improvement over a baseline/reference method, a useful demo, or exceptional creativity.

7. Timeline

Project Proposal — Submit the one-page proposal by the deadline - Week 9 (before Nov 3 23:59).
Final Presentation & Report — Weeks 15–16; exact dates and submission instructions will be announced on the course platform.

8. Where to Find Datasets

Kaggle Datasets — https://www.kaggle.com/datasets
Large collection covering tabular data, computer vision, NLP, time series, and competitions.
UCI Machine Learning Repository — https://archive.ics.uci.edu/datasets

Well-curated classic datasets, especially useful for classification, regression, and tabular ML.

Government Open Data (Taiwan) — https://data.gov.tw/

Public-sector datasets on transportation, environment, health, population, economy, and other real-world topics. Other national/local open-data portals are also acceptable.

Hugging Face Datasets — https://huggingface.co/datasets

Useful for NLP, audio, vision, multimodal, and large-scale datasets; many datasets include dataset cards and previews.

Web Scraping / Public APIs

Teams may collect data from public websites or APIs when appropriate. Clearly document the source, collection procedure, and cleaning steps, and follow the website's terms of service, robots.txt, privacy requirements, and applicable licenses. Prefer official APIs or openly licensed sources when available.

9. Notes

Keep the dataset and computational scope manageable. Interdisciplinary topics are encouraged. If GPU resources are needed, teams are responsible for arranging them (e.g., Google Colab). All team members must understand the problem, model, evaluation, and conclusions; Q&A may be directed to any member.